AI-Powered Phishing Detection & Mid-Strike Infrastructure Analyzer

Project Report & Technical Documentation
Submitted By: Harsh Sandeep Garate
Role: Offensive Cyber Security Intern

1. Executive Summary

Traditional phishing filters rely on static blocklists, which quickly become outdated as threat actors rapidly cycle through new domains and infrastructure. This project addresses that vulnerability by utilizing a behavioral and structural approach. I engineered an AI-driven, multi-vector cybersecurity application that performs real-time machine learning analysis on suspicious URLs, semantic natural language processing (NLP) on communication text, and geospatial tracking of malicious hosting infrastructure.

2. Core Modules & System Architecture

A. Website & Domain Phishing Detection (Machine Learning)

Instead of checking a database, this module extracts mathematical features from a target URL in real-time. It evaluates parameters such as URL length, IP address obfuscation, sub-domain chaining (dot counts), and the presence of suspicious symbols. These features are fed into a trained Scikit-Learn **Decision Tree Classifier**, which calculates the probability of the domain being a phishing vector.

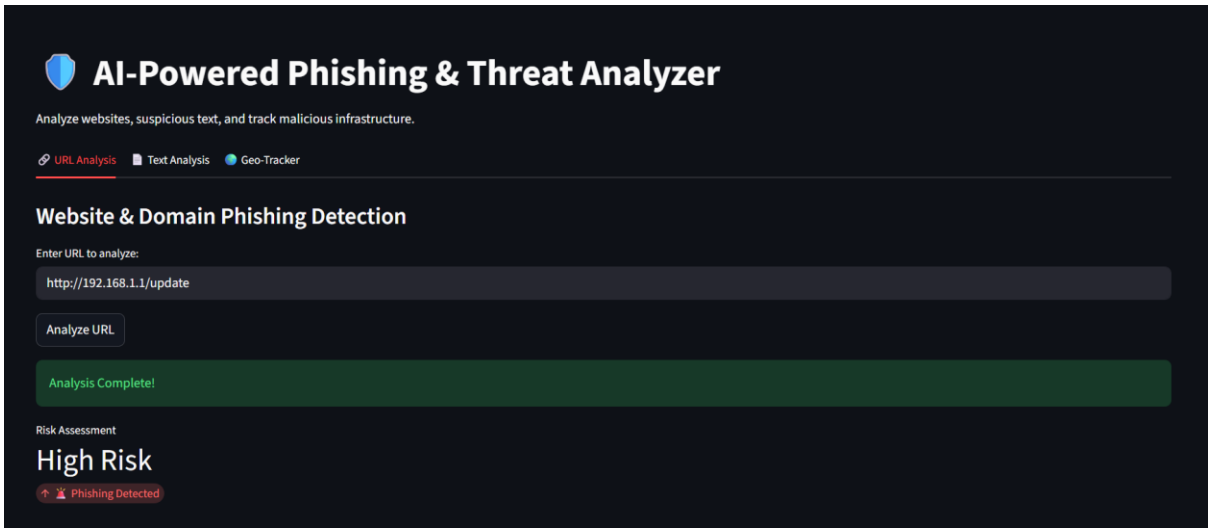


Figure 1: The Decision Tree Classifier successfully identifying high-risk structural patterns in a spoofed URL.

B. Email & Message Phishing Scanner (NLP)

Threat actors frequently use artificial urgency and manipulative vocabulary to execute social engineering attacks. This module uses a Natural Language Processing (NLP) pipeline to analyze the semantics of raw text. Using **TF-IDF vectorization** paired with a **Naive Bayes classification model**, the system flags text containing common phishing hooks, business email compromise (BEC) patterns, and malicious intent.

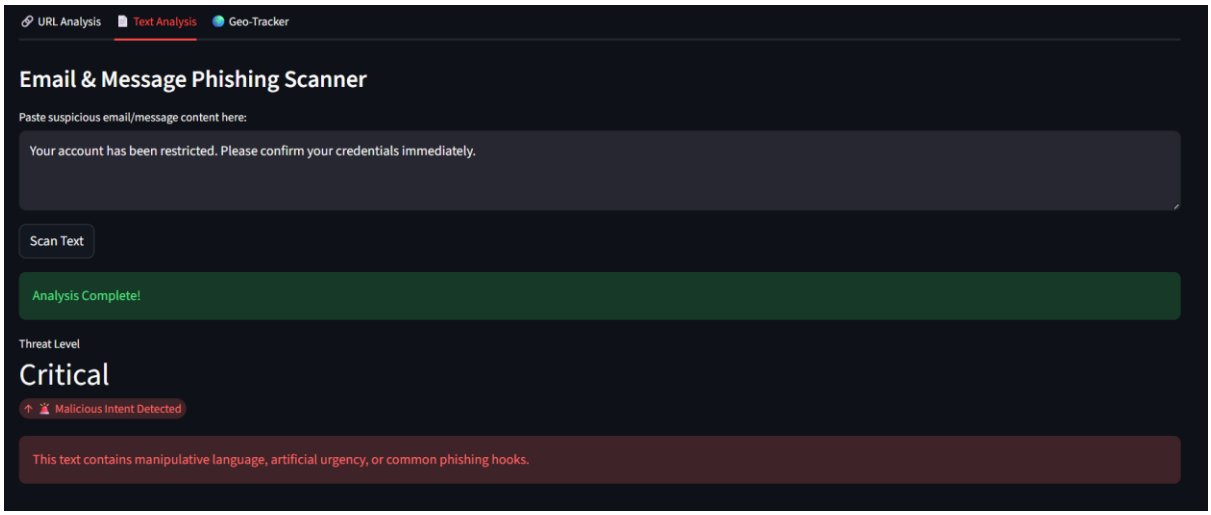


Figure 2: The NLP Naive Bayes model detecting artificial urgency and social engineering tactics in raw communication text.

C. Mid-Strike Infrastructure Tracking (Geo-Intelligence)

Identifying a malicious URL is only the first step in threat intelligence. This module allows security analysts to map the attacker's hosting preferences. It automatically resolves target domains to their underlying IP addresses using network sockets, and then queries a geolocation API to map the Internet Service Provider (ISP), country, and exact city of the server. This data is critical for writing firewall blocking rules and tracking bulletproof hosting providers.

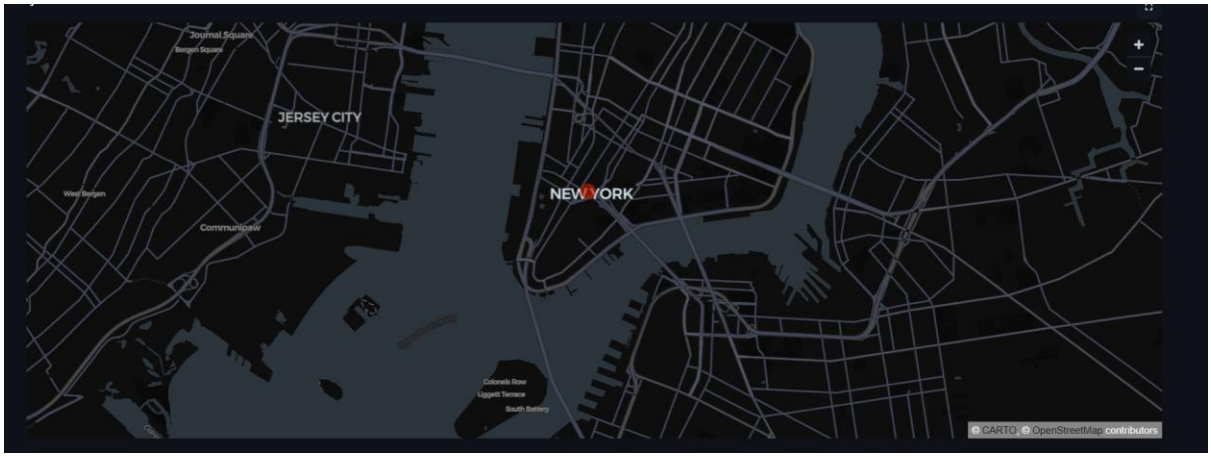
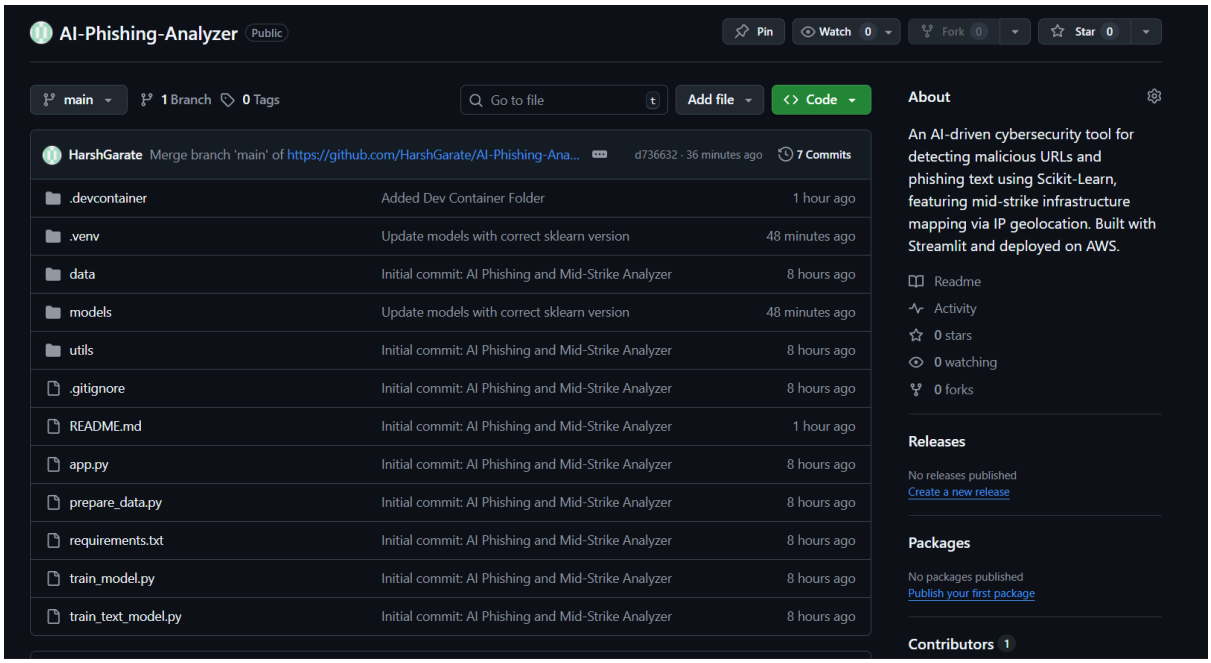


Figure 3.1 & 3.2: Geospatial intelligence mapping the underlying server infrastructure and ISP data of a target domain.

3. Technology Stack & Deployment

The application was built emphasizing speed, accuracy, and seamless deployment.

- **Frontend UI:** Streamlit (Python)
- **Machine Learning:** Scikit-Learn, Pandas, NumPy
- **Data Processing:** Regular Expressions (Regex), TF-IDF Vectorization
- **Network & API Integration:** Python socket library, IP-API
- **CI/CD & Cloud Deployment:** Version-controlled via Git/GitHub and continuously deployed on Streamlit Community Cloud.



4. Conclusion & Future Enhancements

This project successfully demonstrates a full-stack approach to modern threat detection, moving beyond static analysis into AI-driven behavioral prediction. It provides actionable intelligence that can be immediately used in a Security Operations Center (SOC) environment.

Future Scope: Future iterations of this application will focus on containerizing the environment using Docker for deployment on AWS EKS, and integrating premium threat intelligence feeds (such as the VirusTotal API) for cross-referencing file hashes and historical domain reputations.